

Cryptocurrency

S E N D
 + M O R E

 M O N E Y

Character		code
S	—	9
E	—	5
N	—	6
D	—	7
M	—	1
O	—	0
R	—	8
Y	—	2

1) Each letter should be assigned with a value between 0-9

2) Every ~~the~~ letter should be assigned with unique value

1	9			
	1	0		
1	0			

3) The values should be assigned in such a way so that when add up it should give the result as required.

4) For every letter addition the concept of carry to be included.

Sol

So whatever may be the value of 'S' and 'M' that is going to generate a carry, otherwise 'M' will not be generated.

Thus whatever numbers are getting added up the carry is only 1. Hence M has to be 1. Now to generate a carry (M+S) then S=9.

Now $E + \underset{\substack{\uparrow \\ \text{Zero}}}{0} = N$ ~~Now~~ adding '0' to E is leading to a new value. Again any value for E in between 0-9 (except 0, 1, 9) has to be taken, it will lead to

E only as 0 is added. Hence it leads to a carry.

Hence expression

$$E + 1 = N - 1$$

		1	1		
1	9	E	6	7	
	1	0	8	5	
1	0	N	4	2	

Now for $N + R$ the possible combination to generate a carry (except 0, 1, 9) could be

7+8, 6+8, 5+8 -- any one

$$N + R (+1) = E + 10 \text{ --- ii)}$$

Whereas $0 = 0$ (zero) thus

has to get a value to generate carry

As it can not be assured that whether previous addition generates a carry or not

From expressions i) and ii)

$$E + 1 + R (+1) = E + 10$$

$$R (+1) = 9$$

but 9 is already taken

Hence value of $R = 8$ (It can also be other value, the solution is not the unique)

Now $D + E$ again has to generate a carry but 8, 9 are already taken, so 7+6, 5+6 could be the options but 6+4 can not be taken as 0 is already taken.

Let, take $D = 7$ and $E = 5$

So from equation i) $N = 5 + 1 = 6$

* This is not an unique solution but one of the possible solution